CMPE-255 Data Mining Project Report DRAFT

Title - Mental Health & Crisis Signal Mining on Social Media

Cross-source generalization and out-of-distribution safety analysis on Reddit mental-health posts

GitHub Link: https://github.com/NateyLB/CMPE255Project

1. Project Team

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2. Dataset

Two publicly available datasets from Kaggle are used in this project, each playing a clearly distinct role in the pipeline. Both are freely accessible under permissive licenses, require no IRB or data-use agreement, and are loaded directly via the Kaggle API using pandas. The two datasets are not merged into a single training corpus because their label spaces and labeling methodologies differ. Each is used only in the role for which its labels are valid: the Ghoshal Reddit Mental Health Dataset serves as the supervised training corpus, and the Sarkar Sentiment Analysis for Mental Health corpus serves as a held-out external validation corpus that is never seen during training or hyperparameter tuning.

Dataset URLs:

Reddit Mental Health Dataset (Ghoshal):
https://www.kaggle.com/datasets/neelghoshal/reddit-mental-health-data

Sentiment Analysis for Mental Health (Sarkar):
https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health

Dataset	Size	Labels	Role in Project
Reddit Mental Health Dataset (Ghoshal)	~5,957 labeled posts	Five source-subreddit classes: Stress, Depression, Bipolar Disorder, Personality Disorder, Anxiety	Primary 5-class training and in-distribution test set (stratified 80/20 split)
Sentiment Analysis for Mental Health (Sarkar)	~51,000 statements	Seven classes: Normal, Depression, Suicidal, Anxiety, Stress, Bi-Polar, Personality Disorder	External held-out validation set (never seen during training). 5 matched labels → cross-source generalization probe. 2 unseen labels (Suicidal, Normal) → out-of-distribution safety analysis.

3. Abstract - Project Idea

Mental health conditions including depression, anxiety, bipolar disorder, personality disorder, and chronic stress affect hundreds of millions of people worldwide, yet remain significantly underdiagnosed because of stigma, cost, and limited access to care. Social media platforms — particularly Reddit — have become spaces where individuals openly express symptoms and crises, producing large volumes of self-disclosed text suitable for automated analysis. This project investigates whether classical and transformer-based natural language processing techniques can reliably classify Reddit posts into five mental-health conditions, and how those models behave when deployed against an out-of-distribution corpus that contains classes the model was never trained on.

We build a multi-class classification pipeline trained on ~5,957 labeled posts from the Ghoshal Reddit Mental Health Dataset, with five source-subreddit classes: Stress, Depression, Bipolar Disorder, Personality Disorder, and Anxiety. We compare three classical machine-learning approaches (Logistic Regression, Linear SVM, and XGBoost over TF-IDF and engineered features) against a fine-tuned DistilBERT transformer to quantify the gain from contextual representations. Latent Dirichlet Allocation is applied for interpretable topic discovery, and SHAP token-level attributions provide model explainability for the best-performing classifier. A lightweight Streamlit application provides a live inference interface for demonstration.

To probe real-world deployment risk, the best model is evaluated on the Sarkar Sentiment Analysis for Mental Health corpus (~51,000 statements, never seen during training or tuning). The five labels Sarkar shares with our training taxonomy form a cross-source generalization probe; Sarkar's two additional labels — Suicidal and Normal — are unseen during training and form the basis of an out-of-distribution safety analysis. Forced-choice predictions on Suicidal and Normal posts quantify two clinically meaningful failure modes: misrouting of crisis-bearing text (Suicidal posts assigned to a non-suicidal class) and false-positive risk (Normal posts incorrectly flagged as a mental-health condition). This dual analysis converts the label-space mismatch from a limitation into a deliberate experimental contribution.

4. Introduction

4.1 Motivation

The global burden of mental illness is staggering. The World Health Organization estimates that one in eight people worldwide lives with a mental health disorder, yet the majority never receive a diagnosis or treatment. Stigma, cost, and geographic barriers to care mean that many individuals only seek help after a crisis has already escalated.

Social media platforms — Reddit in particular — have become important informal support channels. Users post candidly about depressive episodes, anxiety, bipolar swings, personality-disorder traits, and chronic stress in dedicated subreddits. This produces a large, naturally labeled corpus of self-disclosed mental-health text suitable for automated screening research. At the same time, deploying any such classifier in a setting that affects real people raises immediate questions about generalization across data sources and about behavior on out-of-distribution inputs.

4.2 Objective

This project has three core objectives:

- Build a robust 5-class NLP classification pipeline that distinguishes Stress, Depression, Bipolar Disorder, Personality Disorder, and Anxiety with a high macro-averaged F1 score on a stratified in-distribution test set.
- Compare classical bag-of-words models (TF-IDF + Logistic Regression, Linear SVM, and XGBoost) against a fine-tuned contextual transformer (DistilBERT) to provide an empirical analysis of the accuracy / precision-recall / computational-cost trade-off.
- Quantify cross-source generalization and out-of-distribution behavior using the Sarkar corpus as a held-out validation set: report performance on the five matched labels (cross-source probe) and characterize misrouting and false-positive risk on the two unseen labels Suicidal and Normal (out-of-distribution safety probe).

Throughout, model interpretability is provided through SHAP token-level attributions and LDA topic visualization, supporting auditability of model decisions.

4.3 Technical Challenges

Several challenges shape the design of this pipeline:

- **Class Overlap and Lexical Confusion.** Stress, Depression, and Anxiety language overlap considerably (vocabulary of fatigue, worry, hopelessness, and rumination is shared across all three). Bipolar Disorder and Personality Disorder posts tend to use more clinical-vocabulary phrasing but still share emotional registers with the other classes. Fine-grained 5-way separation is therefore harder than a coarse condition-vs-control task.
- **Class Imbalance.** The Ghoshal source-subreddit classes are roughly balanced (~1,200 per class), which reduces the urgency of resampling, but small absolute counts after the train split limit DistilBERT's effective sample size. We address this by comparing SMOTE oversampling, class-weighted training loss, and unbalanced-baseline conditions in the same evaluation framework.
- **Distributional Shift Across Sources.** Models trained on Ghoshal subreddit-derived labels may not generalize to Sarkar statements, which use a different annotation methodology and a different statement-level distribution. We use Sarkar as an external held-out validation set, never seen during training or hyperparameter tuning, to assess this directly.
- **Out-of-Distribution Behavior.** Sarkar contains two labels (Suicidal and Normal) that the model has never been trained to predict. A naive forced-choice classifier will assign these posts to the closest of its five known classes, which is a deployment failure mode worth quantifying. We treat this as a deliberate analysis rather than as a limitation.
- **Interpretability vs. Accuracy Trade-off.** DistilBERT is expected to outperform classical baselines on raw accuracy but is harder to explain. SHAP token-level attributions and LDA topic clusters bridge this gap and produce auditable evidence behind individual predictions.
- **Subreddit-Style Leakage and Label Proxy Noise.** Class labels in the Ghoshal corpus are derived from source subreddit, not clinical diagnosis. Models can learn subreddit-specific style cues (mod templates, trigger-warning prefixes, in-group vocabulary) instead of mental-health signal — prior work [3] reports very high subreddit-vs-control classification accuracy that is largely an artifact of these surface patterns. We mitigate this by stripping known leakage patterns during preprocessing, by including a stylistic-features-only leakage probe in our experiments, and by reporting per-class precision and recall alongside aggregate macro-F1.

5. Problem statement

Given a Reddit post text P, the primary task is to assign it to one of K = 5 classes:

- C_1 = Stress
- C_2 = Depression

- C_3 = Bipolar Disorder
- C_4 = Personality Disorder
- C_5 = Anxiety

Formally, we seek a classifier $f: X \rightarrow \{C_1, C_2, C_3, C_4, C_5\}$ where X is the space of raw post texts, trained to maximize macro-averaged F1 over a stratified 80/20 train-test split of the Ghoshal corpus. Macro-averaging is used deliberately (rather than accuracy or micro-F1) so that minority-class performance is not masked by the more populous classes.

Two secondary problems follow from the held-out Sarkar corpus:

- **Cross-source generalization (5 matched labels).** Apply f to the subset of Sarkar containing only the five labels matched to our training taxonomy. Report the same metrics as the in-distribution test, and report the gap between in-distribution and cross-source macro-F1 as a measure of distributional robustness.
- **Out-of-distribution safety probe (2 unseen labels).** Apply f to the Sarkar Suicidal and Normal subsets. f is structurally incapable of returning either label, so every prediction is forced into one of the five trained classes. Tabulate the resulting prediction distribution: (i) for Suicidal posts, which trained class absorbs them, with explicit attention to the clinical implication of a suicidal post being routed to Stress, Depression, or another non-crisis-tier class; and (ii) for Normal posts, the false-positive rate per trained class.

Auxiliary goals: discover the latent topical structure of each class via LDA, and explain individual classifications via SHAP token-level attributions.

6. Description of Steps

Step 1 – Data Acquisition & Cleaning

Both datasets are downloaded with the Kaggle API and loaded into pandas, but each is processed in a separate pipeline that respects its own label space and platform. For the Ghoshal corpus, source-subreddit values are mapped to the five target classes per the dataset's documented subreddit-to-condition convention: r/stress → Stress, r/depression → Depression, r/bipolar → Bipolar Disorder, r/BPD → Personality Disorder, and r/anxiety → Anxiety. The exact mapping is committed to utils/common.py as a single source of truth, and any source rows that do not fall into this taxonomy are reported and held aside with the row count documented in the preprocessing log.

Common cleaning operations are applied across both corpora: removal of duplicate posts, null-text rows, HTML tags, URLs, and special characters; lowercasing; and Unicode normalization. To prevent trivial subreddit-style leakage in Ghoshal, we additionally strip subreddit-specific tokens, mod auto-templates, and trigger-warning prefixes (e.g., "TW:", "CW:")

before tokenization. The full leakage-pattern list is constants in the preprocessing notebook so that the stripping step is auditable.

The Sarkar dataset is held strictly as an external validation corpus and is split into two subsets before any model touches it. The five-label matched subset (Stress, Depression, Bipolar Disorder, Personality Disorder, Anxiety) becomes the cross-source generalization probe (EXP-5). The two-label out-of-distribution subset (Suicidal and Normal) is reserved for the OOD safety analysis (EXP-6) — it is never used for training, hyperparameter tuning, threshold selection, or model ranking.

Class imbalance on the primary 5-class task is handled by comparing three conditions in the same evaluation framework: no balancing, class_weight="balanced" training, and SMOTE oversampling on the training split only (never on the test split). A stratified 80/20 train-test split on the Ghoshal corpus, created once and persisted to parquet, ensures label proportions are preserved and that every downstream notebook reads from identical files.

Step 2 – Exploratory Data Analysis (EDA)

EDA characterizes both datasets before modeling. Key analyses include label distribution bar charts, post-length distributions (word count per class), vocabulary size per class, top-n unigram and bigram frequency plots, class-stratified word clouds, and pairwise lexical overlap (Jaccard similarity over the top-100 unigrams per class) to quantify the Stress / Depression / Anxiety overlap noted in §4.3. EDA is repeated on the Sarkar matched-label subset to characterize the cross-source distribution before EXP-5.

Step 3 – Feature Engineering

Four families of features are constructed from the cleaned training corpus:

- **TF-IDF Vectorization.** Unigrams and bigrams, max_features = 50,000, sublinear_tf = True. The vectorizer is fit on the training split only and persisted to artifacts/tfidf_vectorizer.pkl; train and test transforms are saved as sparse matrices.
- **LIWC-Inspired Lexicon Counts.** Per-post counts of words from public emotional-, cognitive-, and social-process category lexicons (NRC Emotion Lexicon or equivalent freely licensed list).
- **Post-Level Surface Statistics.** Word count, sentence count, punctuation density, capitalization ratio, and average word length, used as auxiliary signals for emotional intensity.
- **LDA Topic Probabilities.** After Step 5, the per-post topic probability vector from the optimal LDA model is appended as supplementary features for the EXP-3 ablation.

Step 4 – Model Training & Comparison

Four models are trained on identical train splits and evaluated on the same held-out test set:

- Logistic Regression with TF-IDF features, baseline. C tuned by 5-fold stratified cross-validation over {0.01, 0.1, 1, 10}, L2 penalty.

- Linear SVM with TF-IDF features. C tuned by 5-fold stratified cross-validation.
- XGBoost over TF-IDF concatenated with engineered features (LIWC + post-level + LDA topic probabilities). Hyperparameters tuned with Bayesian search (Optuna or BayesSearchCV) over max_depth, learning_rate, n_estimators, subsample, colsample_bytree, and min_child_weight.
- DistilBERT (distilbert-base-uncased) fine-tuned for up to 3 epochs with AdamW (lr = 2e-5, batch size = 32, weight decay = 0.01, warmup ratio = 0.1) using the HuggingFace Transformers Trainer API. A 10% validation slice is carved from the training split for early stopping on validation macro-F1; the reported number is the best epoch's metric, not necessarily the third.

Each classical model is run under three balancing conditions (no balancing, class-weighted, SMOTE), and metrics for every condition are appended to a shared results/metrics.csv with a fixed schema. A separate stylistic-features-only leakage probe (Logistic Regression on post-level surface statistics with no lexical content) is run as a sanity check against subreddit-style leakage; macro-F1 from this probe is reported alongside the lexical models.

Step 5 – Topic Modeling

Latent Dirichlet Allocation is fit with $k \in \{8, 10, 12, 15, 18, 20\}$ topics. The optimal k is selected by maximizing the C_v coherence score. The fitted model is then used to (a) generate per-post topic probability vectors that feed back into Step 3 for the XGBoost feature ablation, (b) cross-tabulate topic-by-class to identify which latent themes (e.g., isolation, hopelessness, manic energy, identity disturbance, somatic anxiety) co-occur with which trained label, and (c) produce a pyLDAvis HTML visualization for inclusion in the final paper.

Step 6 — Evaluation, Cross-Source Probe, OOD Safety, and Explainability

Each model is evaluated on the held-out Ghoshal test set using macro-averaged F1, weighted F1, per-class precision and recall, ROC-AUC (one-vs-rest, macro-averaged), and a 5×5 confusion matrix. The best macro-F1 model from EXP-1 through EXP-4 is then carried forward to EXP-5 and EXP-6.

EXP-5 (cross-source generalization probe) applies the best model to the Sarkar 5-matched-label subset and reports the same suite of metrics. The macro-F1 gap between the in-distribution test and the cross-source probe is reported as the headline measure of distributional robustness.

EXP-6 (out-of-distribution safety probe) applies the same model to the Sarkar Suicidal and Normal subsets. Two structured analyses are reported: (i) the Suicidal-routing distribution — which of the five trained classes absorbs each Suicidal post, with explicit clinical interpretation of high-Stress or high-Depression routing as misrouting risk, and high-confidence routing as a mis-confidence risk; and (ii) the Normal false-positive distribution — which trained class is most often falsely triggered by Normal text, supplying a deployment-relevant false-positive rate per class.

SHAP TreeExplainer is applied to the XGBoost model to produce global feature importance and force plots for representative correctly-classified and misclassified examples per class. SHAP partition explainer is applied to the DistilBERT model to produce token-level attribution heatmaps over the same examples. Token attributions are also computed on a small sample of OOD Suicidal and Normal posts to surface the lexical drivers behind misrouting and false positives.

A lightweight Streamlit application provides live inference: a user pastes text, sees the predicted class, the per-class confidence vector, and the top SHAP-attributed tokens. A persistent disclaimer on every page makes clear that the demo is for academic demonstration only and is not intended for clinical screening.

7. Related work

7.1 Classical NLP Approaches to Mental Health Detection

Coppersmith et al. [1] established an influential baseline through the CLPsych 2015 shared task on Twitter, demonstrating that TF-IDF-weighted n-gram features combined with standard linear classifiers (SVM, Logistic Regression) can reliably distinguish depression and PTSD posts from controls. Their experimental configuration informs our classical baseline setup (EXP-1 and EXP-2).

7.2 Subreddit-Style Leakage and Weak Supervision

Low et al. [3] document the risk that classifiers trained on subreddit-derived labels can achieve very high accuracy by learning surface stylistic cues (moderation templates, trigger-warning prefixes, in-group vocabulary) rather than genuine mental-health signal. This motivates the explicit preprocessing-stripping step in our Step 1, the per-class precision and recall reporting alongside macro-F1, and the stylistic-features-only leakage probe described in Step 4.

7.3 Deep Learning and Transformer Models

Ji et al. [2] survey machine-learning methods for suicidal-ideation and adjacent mental-health detection and report consistent macro-F1 gains of 5 to 15 percentage points for fine-tuned BERT-family models over classical baselines. Sanh et al. [4] introduce DistilBERT, a 40% smaller and 60% faster distillation of BERT-base that retains roughly 97% of BERT's downstream performance — making it a practical transformer choice for a course project with constrained GPU budget. These results motivate EXP-4 in our experimental design.

7.4 Model Interpretability and SHAP

Lundberg and Lee [5] introduce SHAP, a unified game-theoretic framework for additive feature attribution that subsumes earlier methods (LIME, DeepLIFT) under a single Shapley-value

formulation. SHAP has become a de-facto standard for instance-level interpretability and has been applied to mental-health NLP to surface lexical drivers of model predictions [2]. We adopt SHAP TreeExplainer for XGBoost and SHAP partition explainer for DistilBERT, supporting auditability of individual classifications.

7.5 Out-of-Distribution Behavior in Classifiers

Hendrycks and Gimpel [6] formalize the problem of detecting and characterizing classifier behavior on inputs drawn from outside the training distribution, framing OOD analysis as a deployment-safety question rather than a benchmarking afterthought. Our EXP-6 follows this framing: rather than reporting only matched-label accuracy on Sarkar, we deliberately probe model behavior on Suicidal and Normal subsets that are structurally outside the training taxonomy, and we interpret the resulting prediction distributions as a measure of misrouting and false-positive risk.

8. Solution

Our solution is an end-to-end NLP pipeline with six stages, mapped to the steps in §6 and the experiments in §9:

- **Data integration and cleaning.** Per-corpus pipelines that respect each source's label space. Ghoshal posts are mapped from source subreddit to the five primary classes; Sarkar is partitioned into the 5-matched cross-source probe and the 2-class OOD probe before any model touches it.
- **Feature representation.** Dual-track features: TF-IDF + LIWC + surface statistics (and LDA topic probabilities, added in the EXP-3 ablation) for classical models; raw tokenized sequences for DistilBERT fine-tuning.
- **Multi-model comparison.** Four models trained under identical conditions on the same train-test split, enabling a controlled ablation: LR (baseline) → SVM → XGBoost → DistilBERT (transformer).
- **Topic modeling.** LDA with coherence-based k selection, providing class-level interpretable topic clusters as a supplementary analytical layer and as input features for the XGBoost ablation.
- **Cross-source and OOD evaluation.** EXP-5 measures generalization on the 5 matched Sarkar labels; EXP-6 measures forced-choice routing of unseen Suicidal and Normal posts as a deployment-safety probe.
- **Explainability and deployment.** SHAP attributions on the best in-distribution model and on selected OOD examples; Streamlit app for live demonstration with a clear non-clinical disclaimer.

The pipeline is implemented in Python using pandas, scikit-learn, imbalanced-learn, XGBoost, gensim, pyLDAvis, HuggingFace Transformers and PyTorch, SHAP, and Streamlit. All

notebooks, fitted artifacts, and a shared metrics CSV are version-controlled in the project GitHub repository.

9. List of experiments

All experiments use a single shared experimental protocol:

- Train/Test Split: Stratified 80/20 split on the Ghoshal corpus, persisted to parquet and read identically by every notebook.
- Cross-Validation: 5-fold stratified CV on the training set for hyperparameter tuning.
- Class Imbalance: No-balancing, class_weight="balanced", and SMOTE on the training split only — all three reported.
- Feature Ablation: TF-IDF only versus TF-IDF + engineered features (within EXP-3) to isolate the contribution of hand-crafted features.
- Random Seed: A single RANDOM_SEED = 42 is used in every notebook (numpy, random, torch, transformers) for reproducibility.

Experiment	Model / Method	Feature Input	Expected Macro-F1 (5-class)
EXP-1 (Baseline)	Logistic Regression	TF-IDF (unigrams + bigrams)	0.62 – 0.70
EXP-2	Linear SVM	TF-IDF (unigrams + bigrams)	0.65 – 0.72
EXP-3	XGBoost (Bayesian-tuned)	TF-IDF + engineered (LIWC, surface, LDA)	0.70 – 0.78
EXP-4	DistilBERT (fine-tuned)	Raw token sequences	0.75 – 0.85
EXP-5 (Cross-source)	Best model from EXP-1..4	Sarkar 5-matched-label subset (held out)	Expect 10–20 pt drop vs. EXP-4
EXP-6 (OOD safety probe)	Best model from EXP-1..4	Sarkar Suicidal + Normal (unseen during training)	Qualitative + structured routing analysis
LP-Probe (Leakage)	Logistic Regression	Post-level surface stats only (no lexical content)	Should be ≤ 0.40 if leakage is controlled

10. Proposed Evaluation Metrics and Justification

Metric	Justification
Macro-averaged F1	Treats each of the five trained classes equally regardless of support. Primary ranking metric for selecting the best model from EXP-1..4.
Per-class Precision / Recall	Precision controls false alarms; recall controls missed cases. Both are clinically meaningful for the three lexically overlapping classes (Stress, Depression, Anxiety) and for the two more clinical classes (Bipolar Disorder, Personality Disorder).
ROC-AUC (per class, one-vs-rest, macro-averaged)	Measures separability between each class and the rest, providing a threshold-independent ranking metric to complement macro-F1.
Confusion Matrix	Provides a full 5×5 error breakdown showing which classes are most often confused. Informs feature-engineering and preprocessing follow-ups.
Cross-source Macro-F1 Gap (EXP-5)	The drop from in-distribution macro-F1 to cross-source macro-F1 on the 5 matched Sarkar labels — a direct measure of distributional robustness.
OOD Routing Distribution (EXP-6)	Tabulates which trained class absorbs each unseen-class input. Reports two clinically meaningful figures: the Suicidal-misrouting distribution (where do crisis-bearing posts go?) and the per-class Normal false-positive rate (which class is most often falsely flagged?).
LDA Coherence (C_v)	Selects the optimal number of latent topics k by measuring the semantic coherence of top words per topic.
SHAP Token Importance	Provides instance-level interpretability by attributing predictions to individual tokens — essential for auditability in any deployment-adjacent context.

Primary ranking metric: macro-averaged F1 on the in-distribution test set. Secondary headline: cross-source macro-F1 gap (EXP-5). Tertiary headlines: Suicidal-routing distribution and Normal

false-positive rate (EXP-6). The best macro-F1 model from EXP-1..4 is the target for SHAP analysis and for the EXP-5 / EXP-6 probes.

11. Results

11.1 Exploratory Data Analysis – Initial Findings

Both datasets have been loaded and explored. The full EDA notebook is available in the GitHub repository at notebooks/01_EDA.ipynb.

Dataset sizes (after dropping null-text rows):

- Reddit Mental Health (Ghoshal): 5,957 posts across the five source-subreddit classes (Stress, Depression, Bipolar Disorder, Personality Disorder, Anxiety), roughly balanced at ~1,200 posts per class.
- Sentiment Analysis for Mental Health (Sarkar): ~51,000 statements across seven classes (Normal, Depression, Suicidal, Anxiety, Stress, Bi-Polar, Personality Disorder). For our cross-source probe (EXP-5) we filter to the five labels matched with our training taxonomy. The two unseen labels (Suicidal, Normal) are reserved for the OOD safety analysis (EXP-6).

Key initial observations:

- The Ghoshal corpus is approximately balanced across all five classes, which reduces the urgency of aggressive resampling on the primary task. SMOTE will still be evaluated as one of three balancing conditions for completeness.
- Initial unigram and bigram analysis on Ghoshal shows substantial lexical overlap between Stress, Depression, and Anxiety classes — common high-frequency tokens such as "feel," "people," "time," "work," and "life" appear across categories. This confirms the lexical-confusion challenge documented in §4.3 and motivates the use of contextual embeddings (DistilBERT) alongside TF-IDF baselines.
- Bipolar Disorder and Personality Disorder posts have visibly longer average post lengths and a higher density of clinical-vocabulary tokens (e.g., explicit references to medication, therapy modalities, episode descriptions) — these may be the easier-to-separate classes in the supervised setting.
- Sarkar EDA on the matched 5-label subset is in progress. Initial loading confirms that all five matched labels are well-represented, with sufficient row counts to support the cross-source probe. The distribution of post lengths in Sarkar differs from Ghoshal — Sarkar statements are noticeably shorter on average — which is a known source of cross-source distributional shift and is expected to contribute to the EXP-5 macro-F1 gap.

Sarkar Suicidal and Normal subsets have been isolated and are under access lock until EXP-5 reports. They will not be touched by any training, tuning, or threshold-selection step.

11.2 Baseline Model (Placeholder)

Baseline runs (Logistic Regression and Linear SVM with TF-IDF features on the Ghoshal 5-class corpus) are currently in progress. Preliminary macro-F1, per-class precision/recall, ROC-AUC, and 5×5 confusion matrices will be reported in the Check-in 3 deliverable, alongside the XGBoost and DistilBERT comparisons described in §9. The leakage-probe macro-F1 will be reported alongside the baseline numbers as a sanity check on subreddit-style leakage.

12. Conclusions

Pending complete model runs, we expect the following directional outcomes — these are working hypotheses that the final report will validate or refute against measured numbers.

On the in-distribution Ghoshal test set, we expect DistilBERT to outperform classical baselines (Logistic Regression, SVM, XGBoost) by approximately 5 to 12 macro-F1 points. The headroom is bounded by the relatively small training set (~4,800 posts after the 80/20 split) — published transformer gains on much larger corpora may not transfer fully at this scale.

On the EXP-5 cross-source probe (Sarkar, 5 matched labels), we expect a notable macro-F1 drop compared to the in-distribution test — likely in the 10 to 20 percentage-point range — driven by platform and labeling-methodology differences. We do not interpret this drop as a failure of the model, but as a measured property of subreddit-derived weak supervision, and we present the gap as the central robustness finding of the project.

On the EXP-6 OOD safety probe, we expect Suicidal posts to be predominantly routed to Depression and to a lesser extent to Stress, with relatively low routing to Bipolar Disorder, Personality Disorder, and Anxiety. We expect Normal posts to spread more uniformly across the five trained classes, with the highest false-positive rate on Stress. Both outcomes have direct deployment implications: high-confidence Suicidal-as-Depression routing represents clinical mis-prioritization, and a high Normal-as-Stress false-positive rate represents an everyday-noise alarm risk. Quantifying both as explicit numbers is, in our view, more useful than reporting only matched-label accuracy.

Per-class precision and recall — rather than aggregate macro-F1 alone — will be the operative metrics for assessing whether the models distinguish the lexically overlapping Stress, Depression, and Anxiety classes versus learning subreddit-style cues. SHAP token attributions and LDA topic clusters are expected to surface interpretable lexical and topical patterns for each class (e.g., somatic-anxiety vocabulary clustering with Anxiety, manic-energy vocabulary with Bipolar Disorder, identity-disturbance vocabulary with Personality Disorder), and to surface specific tokens responsible for the most consequential OOD routing decisions.

13. Limitations

- **Cross-source heterogeneity.** Ghoshal and Sarkar differ in post format (full Reddit posts vs. shorter statements), label granularity (5 vs. 7), and labeling methodology (subreddit-derived vs. mixed). We do not merge them into a single training set; cross-source numbers are reported as deliberate generalization probes, not as in-distribution performance.
- **Self-report bias and label-proxy noise.** Reddit posts are self-reported and unverified by clinical professionals. Ghoshal labels are derived from subreddit membership rather than clinical diagnosis, introducing label noise and the risk of subreddit-style leakage. Mitigations are described in §6 Step 1 and §7.2.
- **Out-of-distribution probe is forced-choice.** The EXP-6 analysis treats every Suicidal and Normal post as a forced-choice prediction over the five trained classes. We do not implement OOD detection itself (e.g., softmax entropy thresholding, energy-based scoring); doing so would extend the project beyond its current scope and is listed as future work.
- **Temporal generalization.** Mental-health language on social media changes over time (post-COVID shifts, slang turnover, platform moderation changes). Models trained on historical data may underperform on future posts.
- **GPU access constraints.** DistilBERT fine-tuning requires a GPU. We plan to use Kaggle Notebooks (free P100/T4) as the canonical training environment with Google Colab as a development backup. Limited GPU time may constrain the number of hyperparameter search iterations we can run for EXP-4.
- **Class overlap.** Stress, Depression, and Anxiety language overlaps significantly. Aggregate macro-F1 may understate or overstate model quality if the overlap is unevenly distributed across classes; this is why per-class precision and recall are first-class outputs of every experiment.
- **Ethical scope.** This project does not deploy predictions in any clinical context. The Streamlit demo is for academic demonstration only and includes a persistent disclaimer to that effect. The OOD analysis reports failure modes for awareness and for paper discussion, not as endorsement of clinical-decision use.
- **Training-set size for transformer fine-tuning.** Approximately 4,800 training posts after the 80/20 split is small for transformer fine-tuning by modern standards. Reported DistilBERT macro-F1 may need recalibration after final cross-validation results are available; we use early stopping on a held-out validation slice to mitigate overfitting.

14. Future work

- Cross-platform generalization: extend evaluation to Twitter or Facebook mental-health corpora to test platform-level robustness beyond the Reddit-only training distribution.
- True OOD detection: implement softmax-entropy thresholding or energy-based OOD scoring so that unseen-class inputs can be flagged as "unknown" rather than forced into one of the five trained classes.
- Domain-specialized transformers: evaluate MentalBERT or mental-RoBERTa, pre-trained on mental-health text, as alternatives to general-domain DistilBERT — likely the highest-leverage extension after the present project.
- Longitudinal modeling: model temporal trajectories of individual users (post-sequence modeling with RNNs or temporal transformers) to capture clinical course rather than single-post snapshots.
- Multimodal behavioral signals: incorporate user engagement metadata (post frequency, upvote ratio, reply depth) as auxiliary features.
- Clinical validation: collaborate with mental-health professionals to validate LDA topic clusters and SHAP-identified keywords against established symptom criteria (DSM-5).
- Active learning: implement a feedback loop where flagged predictions are reviewed and corrected to improve the model over time and reduce labeling cost.

15. Contributions

Team Member	Role	Key Responsibilities
Jesigga Sigurdardottir	Data Engineer	Data acquisition (Kaggle API), per-corpus preprocessing pipelines, leakage-pattern stripping, stratified 80/20 split, SMOTE setup. EDA on Ghoshal and Sarkar (matched and OOD subsets). Maintains preprocessing report and label-mapping dictionary.

Ravikumar Komandur Narayanan	NLP Feature Engineer	TF-IDF vectorization, n-gram analysis, LIWC-inspired lexicon counts, post-level surface statistics. LDA topic modeling, coherence-based k selection, pyLDAvis visualization, topic-by-class cross-tabulation. Word clouds and per-class lexical fingerprints.
Gunanidhi Ramakrishnan	ML Modeling Lead	Logistic Regression, Linear SVM, and XGBoost training and tuning (5-fold CV, Bayesian search). Owns utils/common.py (shared metrics, label map, random seed). Cross-source EXP-5 and OOD EXP-6 evaluation notebooks; final comparison and confusion-matrix figures.
Nathan Howland	Deep Learning & Deployment	DistilBERT fine-tuning (HuggingFace Transformers), early-stopping configuration, class-weighted training. SHAP TreeExplainer (XGBoost) and SHAP partition explainer (DistilBERT) figures, including OOD-routing attributions. Streamlit demo application; lead on paper write-up and slide deck preparation.

16. References

[1] Coppersmith, G., Dredze, M., Harman, C., Hollingshead, K., & Mitchell, M. (2015). CLPsych 2015 shared task: Depression and PTSD on Twitter. Proceedings of the 2nd Workshop on Computational Linguistics and Clinical Psychology (CLPsych), pp. 31–39. https://aclanthology.org/W15-1204/

[2] Ji, S., Pan, S., Li, X., Cambria, E., Long, G., & Huang, Z. (2021). Suicidal ideation detection: A review of machine learning methods and applications. IEEE Transactions on Computational Social Systems, 8(1), 214–226. https://doi.org/10.1109/TCSS.2020.3021467

[3] Low, D. M., Rumker, L., Torous, J., Cecchi, G., Ghosh, S. S., & Talkar, T. (2020). Natural Language Processing Reveals Vulnerable Mental Health Support Groups and Heightened

Health Anxiety on Reddit During COVID-19: Observational Study. Journal of Medical Internet Research, 22(10), e22635. https://doi.org/10.2196/22635

[4] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108. https://arxiv.org/abs/1910.01108

[5] Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems (NeurIPS), 30, pp. 4765–4774. https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[6] Hendrycks, D., & Gimpel, K. (2017). A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. International Conference on Learning Representations (ICLR). https://arxiv.org/abs/1610.02136